

Step 1: add ip address for interfaces and add DHCP server in R1

```

R1(config)#ip route 192.168.2.0 255.255.255.0 172.16.0.2
R1(config)#ip dhcp pool hr
R1(dhcp-config)#network 50.0.0.0 255.0.0.0
R1(dhcp-config)#default-router 50.0.0.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 1 2
R1(dhcp-config)#exit
R1(config)#ip excluded-address 50.0.0.1 50.0.0.50
      ^
% Invalid input detected at '^' marker.

R1(config)#ip excluded-address 50.0.0.1 50.0.0.50
*Aug  3 03:04:21.517: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R1(config)#ip dhcp excluded-address 50.0.0.1 50.0.0.50
R1(config)#
R1(config)#ip dhcp pool sales
R1(dhcp-config)#network 192.168.2.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.2.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#lease 1 2
R1(dhcp-config)#exi

```

Step 2: Add interfaces and Relay-agent in R2

```

Gateway of last resort is not set

S    50.0.0.0/8 [1/0] via 172.16.0.1
S    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.16.0.0/16 is directly connected, Ethernet0/1
L    172.16.0.2/32 is directly connected, Ethernet0/1
S    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, Ethernet0/0
L    192.168.2.1/32 is directly connected, Ethernet0/0
R2#
*Aug  3 03:11:01.602: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#int eth 0/1
R2(config-if)#exit
R2(config)#int eth 0/0
R2(config-if)#ip helper-address 172.16.0.1
R2(config-if)#

```

Step 3 : fetch ip address from dhcp server using relay agent

```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 192.168.2.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 192.168.2.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.2.1

C:\Documents and Settings\user>
```

Step 4: Getting Ip address from R1's Dhcp server

```
Welcome to Virtual PC Simulator, version 1.3 (0.8.1)
Dedicated to Daling.
Build time: May  7 2022 15:27:29
Copyright (c) 2007-2015, Paul Meng (mirnshi@gmail.com)
Copyright (c) 2021, Alain Degreffe (alain.degreffe@eve-ng.net)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
Modified version for EVE-NG.

Press '?' to get help.

Executing the startup file

VPC-2> ip dhcp
ODORA IP 50.0.0.51/8 GW 50.0.0.1

VPC-2> █
```